

Feasibility Study

Project: Library Management System (LMS)

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1. Technical Feasibility

Question: Can we build it with available technology and skills?

- The system is a standard web application (login, forms, database, reports). These are mature technologies.
- Proposed stack: HTML/CSS/JavaScript front end, and a relational database such as MySQL.
- Hardware: a server or cloud hosting, and optionally barcode scanners for faster check-in and check-out.
- Team skills: web development and database basics are enough for this scope.
- Technical risks: wrong data when moving old records, and system downtime. Mitigation: data checks during import, regular backups.

Conclusion: Technically feasible. Risk is low.

2. Operational Feasibility

Question: Will the users accept and use the system?

- Librarians want less paperwork and members want faster search and borrowing, so both groups benefit.
- The interface is simple (search, borrow, return, reports), so little training is needed.
- Possible resistance: staff used to paper registers. Mitigation: short training and a user guide.

Conclusion: Operationally feasible.

3. Schedule Feasibility

Question: Can it be delivered on time?

- The scope is small and clear (no e-books, no online payment).
- Estimated duration: about 3-4 months from analysis to deployment.

Conclusion: Feasible if the scope stays fixed.

4. Economic Feasibility

All amounts are in USD and are estimates (replace them with real quotes when available).

Time horizon: 5 years. Discount rate: 10%.

Excel sheet:

Cost-Benefit Analysis - Library Management System

All amounts in USD (estimates)

Category	Description	Estimated Cost	Benefits	Estimated Annual Value
Software	Development (in-house) or purchase of an LMS license	\$2,000 - \$5,000	Centralized, automated system reduces errors and manual record keeping	\$3,000 saved in staff time
Hardware	Server, barcode scanners (optional), backup drives	\$1,500 - \$3,000	Faster book check-in/out; reduced manual work	\$2,500 saved in staff time
Training	Staff and member training sessions	\$500 - \$1,000	Higher productivity; smooth transition and efficient use of the system	\$1,000 gain
Maintenance	Annual software updates, bug fixes, server hosting	\$800 - \$1,200	Ensures system reliability and security; fewer lost books and record errors	\$2,000 saved
Administration	Current manual system (registers, spreadsheets, paper forms, printing)	\$1,200 annually (current cost)	Eliminated with the LMS, so the library becomes paperless	\$1,200 saved
User Experience	Manual searches waste time for members	-	Faster access to books, improved satisfaction	Intangible but high

Summary

Total Estimated Costs (first year):	\$4,800 - \$10,200
Total Estimated Annual Benefits/Savings:	\$8,000 - \$10,000+
Payback Period:	About 6 to 13 months (around 1 year)

6. Final Recommendation

The project is technically, operationally, schedule-wise and economically feasible (positive NPV of \$26,480, ROI 322%, break-even in about 9 months). Recommendation: proceed with the project.